

## OSAN AB, Korea, Republic of

WMO: 471220

Lat: 37.091N

Lon: 127.030E

Elev: 38

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 8.4        | 12.1 | -1.9                          | 5.0 | 16.4 | 1.7 | 6.0 | 18.6 | 21.6                  | 28.5 | 19.4 | 28.1 | 4.7                   | 70   | 0.439 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 14.2                   | 92.2            | 78.8 | 89.6 | 77.7 | 87.4 | 76.4 | 82.0                | 86.2 | 80.4 | 85.6 | 79.1 | 84.2 | 7.6                  | 280  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 80.9                            | 160.8 | 83.1 | 79.1 | 151.2 | 82.2 | 77.4 | 142.9 | 81.2 | 46.0          | 88.5 | 44.2 | 85.1 | 42.8 | 83.6 | 89.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| (4) 18.8          | 16.3 | 14.1 | 2.9                        | 94.9 | 4.5                | 2.7 | -0.4                                               | 96.8 | -3.0       | 98.4 | -5.5       | 99.9 | -8.8       | 101.9 |    |    |
| (5)               |      |      | 2.1                        | 84.1 | 4.1                | 2.3 | -0.9                                               | 85.7 | -3.3       | 87.0 | -5.6       | 88.3 | -8.6       | 89.9  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 54.6   | 27.3  | 31.9  | 41.8  | 53.5  | 64.0  | 72.5  | 78.6  | 79.5  | 70.3  | 58.0  | 44.5  | 31.5  |  |  |
|                                                                        | DBStd       | 19.08  | 7.37  | 7.13  | 6.48  | 6.45  | 4.84  | 3.80  | 4.10  | 4.64  | 4.90  | 6.27  | 7.63  | 7.56  |  |  |
|                                                                        | HDD50       | 2295   | 703   | 507   | 264   | 35    | 0     | 0     | 0     | 0     | 0     | 13    | 198   | 575   |  |  |
|                                                                        | HDD65       | 5129   | 1168  | 926   | 719   | 348   | 76    | 1     | 0     | 0     | 0     | 226   | 615   | 1039  |  |  |
|                                                                        | CDD50       | 3967   | 0     | 1     | 11    | 140   | 435   | 675   | 886   | 916   | 609   | 261   | 33    | 0     |  |  |
|                                                                        | CDD65       | 1328   | 0     | 0     | 0     | 4     | 47    | 226   | 421   | 451   | 170   | 9     | 0     | 0     |  |  |
|                                                                        | CDH74       | 11083  | 0     | 0     | 0     | 60    | 451   | 1541  | 3662  | 4245  | 1054  | 69    | 1     | 0     |  |  |
|                                                                        | CDH80       | 3626   | 0     | 0     | 0     | 6     | 80    | 406   | 1265  | 1640  | 227   | 2     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 5.7    | 5.5   | 5.9   | 6.9   | 6.7   | 6.2   | 5.4   | 5.7   | 5.3   | 4.7   | 5.0   | 5.4   | 5.4   |  |  |
|                                                                        | PrecAvg     | 54.0   | 0.7   | 1.1   | 1.7   | 3.2   | 3.7   | 5.7   | 15.2  | 11.4  | 6.6   | 1.9   | 1.7   | 1.0   |  |  |
| (15) Precipitation                                                     | PrecMax     | 84.2   | 2.5   | 3.1   | 5.1   | 7.4   | 9.4   | 17.0  | 34.5  | 34.6  | 23.1  | 7.3   | 5.0   | 2.3   |  |  |
|                                                                        | PrecMin     | 31.6   | 0.0   | 0.0   | 0.2   | 0.8   | 0.5   | 1.1   | 5.0   | 2.4   | 0.8   | 0.0   | 0.3   | 0.2   |  |  |
|                                                                        | PrecStd     | 12.3   | 0.6   | 0.8   | 1.1   | 1.7   | 1.8   | 3.7   | 6.9   | 7.5   | 5.8   | 1.5   | 1.0   | 0.6   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 50.4  | 59.1  | 68.1  | 80.4  | 86.1  | 89.7  | 95.1  | 96.7  | 87.9  | 79.2  | 69.6  | 55.6  |  |  |
|                                                                        |             | MCWB   | 44.0  | 51.0  | 53.7  | 63.5  | 63.2  | 70.7  | 78.3  | 79.7  | 73.7  | 65.1  | 59.0  | 48.6  |  |  |
|                                                                        | 2%          | DB     | 46.1  | 52.2  | 62.8  | 75.1  | 82.2  | 86.4  | 91.6  | 93.3  | 85.2  | 75.5  | 64.7  | 50.4  |  |  |
|                                                                        |             | MCWB   | 40.5  | 43.9  | 50.4  | 58.4  | 64.3  | 70.2  | 78.9  | 79.2  | 72.3  | 63.4  | 55.8  | 44.6  |  |  |
|                                                                        | 5%          | DB     | 42.5  | 48.2  | 58.9  | 71.3  | 78.8  | 84.4  | 89.3  | 91.1  | 82.5  | 73.0  | 61.2  | 47.0  |  |  |
|                                                                        |             | MCWB   | 37.5  | 40.8  | 48.1  | 56.3  | 63.0  | 69.7  | 78.1  | 79.0  | 71.0  | 62.1  | 53.7  | 42.5  |  |  |
|                                                                        | 10%         | DB     | 39.2  | 44.4  | 55.1  | 67.7  | 75.6  | 82.3  | 86.3  | 88.1  | 80.3  | 69.9  | 57.5  | 44.2  |  |  |
|                                                                        |             | MCWB   | 35.3  | 38.6  | 45.9  | 54.6  | 62.2  | 69.4  | 77.1  | 78.2  | 69.5  | 60.9  | 51.8  | 40.2  |  |  |
|                                                                        | 0.4%        | WB     | 45.4  | 53.3  | 57.6  | 66.7  | 72.2  | 79.0  | 84.1  | 85.9  | 80.2  | 70.2  | 62.9  | 50.2  |  |  |
|                                                                        |             | MCDB   | 48.7  | 56.7  | 62.6  | 74.0  | 76.5  | 81.1  | 87.2  | 87.7  | 83.1  | 74.4  | 65.8  | 53.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 41.3  | 45.3  | 53.4  | 62.1  | 69.1  | 75.8  | 81.5  | 82.6  | 76.7  | 67.3  | 58.5  | 46.2  |  |  |
|                                                                        |             | MCDB   | 44.4  | 50.2  | 59.4  | 70.6  | 74.7  | 79.5  | 85.6  | 86.7  | 80.2  | 71.4  | 62.3  | 49.0  |  |  |
|                                                                        | 5%          | WB     | 38.3  | 41.9  | 50.0  | 59.0  | 66.8  | 74.0  | 80.2  | 80.9  | 74.6  | 64.8  | 55.4  | 43.1  |  |  |
|                                                                        |             | MCDB   | 41.0  | 46.6  | 55.7  | 66.7  | 73.5  | 78.7  | 84.9  | 86.5  | 78.6  | 69.3  | 59.3  | 46.3  |  |  |
|                                                                        | 10%         | WB     | 35.6  | 39.3  | 46.9  | 56.3  | 64.8  | 72.3  | 79.0  | 79.7  | 72.4  | 62.7  | 52.4  | 40.4  |  |  |
|                                                                        |             | MCDB   | 38.1  | 43.1  | 53.1  | 63.9  | 71.7  | 77.4  | 83.8  | 85.4  | 76.7  | 67.7  | 56.6  | 43.5  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 19.0  | 20.4  | 21.6  | 23.2  | 21.3  | 17.9  | 13.5  | 14.2  | 17.7  | 21.5  | 20.0  | 18.6  |  |  |
|                                                                        |             | MCDBR  | 20.6  | 25.4  | 28.0  | 29.0  | 26.5  | 21.9  | 17.1  | 17.6  | 19.8  | 23.5  | 23.7  | 21.8  |  |  |
|                                                                        |             | MCWBR  | 15.3  | 17.6  | 16.6  | 14.7  | 11.5  | 9.3   | 7.2   | 7.3   | 9.3   | 12.6  | 15.8  | 16.5  |  |  |
|                                                                        |             | MCDBR  | 18.6  | 21.7  | 23.7  | 23.5  | 20.3  | 16.6  | 13.9  | 14.7  | 15.6  | 19.5  | 20.1  | 20.3  |  |  |
|                                                                        | 5% WB       | MCWBR  | 15.1  | 16.5  | 16.3  | 14.1  | 11.7  | 9.6   | 7.7   | 7.8   | 9.1   | 13.0  | 15.4  | 16.5  |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.443 | 0.491 | 0.574 | 0.611 | 0.625 | 0.697 | 0.658 | 0.582 | 0.509 | 0.497 | 0.480 | 0.439 |  |  |
|                                                                        | taud        |        | 1.942 | 1.856 | 1.710 | 1.677 | 1.697 | 1.618 | 1.747 | 1.925 | 2.051 | 1.958 | 1.943 | 1.967 |  |  |
|                                                                        | Ebn at Noon |        | 220   | 226   | 220   | 222   | 222   | 206   | 213   | 227   | 236   | 222   | 206   | 210   |  |  |
|                                                                        | Edh at Noon |        | 45    | 55    | 70    | 76    | 76    | 82    | 72    | 59    | 49    | 49    | 44    | 41    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 785   | 1033  | 1360  | 1581  | 1773  | 1643  | 1315  | 1387  | 1304  | 1140  | 781   | 691   |  |  |
|                                                                        | RadStd      |        | 48    | 84    | 136   | 117   | 164   | 161   | 195   | 199   | 142   | 84    | 82    | 49    |  |  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (48 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +169  | +90   |

Nomenclature: See separate page